

Amr Muhammad Mahdi Alnus

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[LinkedIn](#) — [GitHub](#)

Education

Alexandria University

Oct 2021 – Aug 2026

B.Sc. in Electronics and Communication Engineering

- GPA: 3.35 / 4.0

Graduation Project - E Learning Platform

- Integrated Agora SDK for real-time video conferencing functionality.
- Integrated Google Calendar API for automated scheduling and event management.
- Refactored backend to follow clean architecture principles, improving scalability and maintainability.
- Developed unit tests to enhance code reliability and ensure system stability.

Projects

Workout Tracker

[GitHub](#)

- Built a scalable backend for workout tracking using Node.js, Express, TypeScript, Prisma, and PostgreSQL, supporting JWT authentication and full CRUD operations
- Implemented unit and integration tests with Jest and Supertest, achieving 81% code coverage
- Performed load testing with k6, simulating up to 500 concurrent users and with lower than 0.05% error rate
- Monitored and improved performance using Prometheus, Grafana, and PM2 clustering; identified p95 latency bottlenecks (1.3–1.45s under peak load)
- Technologies: Node.js, Express.js, TypeScript, Prisma, Jest, Grafana, Prometheus, K2

E-commerce API

[GitHub](#)

- Built a backend API for an e-commerce platform with Stripe payment integration
- Designed scalable data models with MongoDB
- Technologies: Node.js, Express.js, MongoDB, Stripe, Jest

Caching Proxy Server

[GitHub](#)

- Built an HTTP caching proxy reducing repeated request latency
- Implemented cache validation and request forwarding logic
- Technologies: Node.js, Networking

URL Shortening Service

[GitHub](#)

- Designed and implemented a backend service exposing REST APIs for URL shortening
- Technologies: Node.js, Prisma, PostgreSQL, Jest

Discord Bot

[GitHub](#)

- Developed automation features handling multiple Discord workflows
- Integrated MongoDB with Mongoose for persistent storage
- Designed event-driven architecture using Discord.js
- Technologies: Node.js, MongoDB, Discord.js

Image Processing Service

[GitHub](#)

- Implemented backend for user authentication and image processing
- Image upload and transformation pipelines using Multer and Cloudinary
- Technologies: Node.js, TypeScript, Express, Prisma, SQLite, Cloudinary, Redis, RabbitMQ, Jest

Movies App (Flutter)

GitHub

- Developed profile and search screens with Clean Architecture and Cubit
- Integrated SQLite to handle user history
- Technologies: Flutter, SQLite, Clean Architecture, Cubit

Events App (Flutter)

GitHub

- Implemented Firebase Authentication and Google Maps integration
- Used Provider state management for theming and localization
- Technologies: Flutter, Firebase, Provider, Google Maps

Training & Certifications

Holberton School (ALX)

2023 – 2024

Software Engineering — 12-Month Program

- Completed 1,000+ hours of backend-focused training in Bash, C, Python, JavaScript, and TypeScript
- Designed and implemented RESTful APIs serving multiple resources using Flask and SQLAlchemy
- Worked extensively with relational and NoSQL databases including MySQL, MongoDB, and Redis

Route Academy

Jul 2025 – Dec 2025

Flutter Cross-Platform Application Development Diploma

- Developed 4+ cross-platform mobile applications using Flutter and Dart
- Applied SOLID principles and Clean Architecture to reduce code coupling and improve maintainability
- Integrated RESTful APIs and implemented reusable UI components across multiple screens
- Used Firebase Authentication and Firestore for real-time data synchronization

Information Technology Institute (ITI)

Jul 2025 – Aug 2025

Linux System Administration

Extracurricular Activities

IEEE SCSS Alexandria Branch — Head of Software Engineering

Feb 2025 – Aug 2025

- Interviewed and evaluated 30+ candidates for technical roles
- Designed training plans and mentored software engineering members

Alex Eagles — Software Engineer

Sep 2024 – Mar 2025

- Developed autonomous drone control software using Python and DroneKit
- Improved command-response latency between drone and ground station during missions

Femto Science — English Instructor

Oct 2022 – Jul 2023

- Delivered online lectures and evaluated student assignments

Skills

Languages: Arabic (Native), English (Advanced), German (Beginner)

Programming Languages: C, C++, Python, Java, JavaScript, TypeScript, Dart, SQL, Bash

Backend & Web: REST APIs, Node.js, Express.js, Prisma, JWT, Jest, Supertest, RBAC, SOLID, Design Patterns

Frontend & Mobile: Flutter, Clean Architecture, Cubit, Provider, Firebase

Databases: PostgreSQL, MySQL, MongoDB, Redis, SQLite

DevOps & Tools: Git, Linux, Docker, Postman, Cloudinary